

## • review

### • review on linear and non-linear Functions

- a linear Function, the highest power of "x" is one if a Function includes  $x^2$  its no longer linear but rather a quadratic Function

example of a linear Function:  $2x+3$

example of a non-linear Function:  $2x^2+3$

### • review on derivatives

- Product rule (For two terms being multiplied):

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

- Quotient rule (For anything that has division):

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

### • integral review

you only need a "+ C" at the end of the integral if its undefined or looks like the "∫"

you do not need a "+ C" if your integral is defined or looks like this " $\int_a^b$ "

# equation of line

- equation of a line in a 3d space is determined by Point  $P_0(x_0, y_0, z_0)$  • usually given

$P(x_0, y_0, z_0)$  - Point on the Plane

$\vec{r} = \langle a, b, c \rangle$  - Perpendicular to the Plane

- Vector Form

$$\langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

- Vector equation

$$\vec{r}(t) = \vec{r}_0 + t\vec{v}$$

•  $\vec{r}_0$  is a Position Vector specific Point on the line

•  $\vec{v}$  is the direction Vector of the line

•  $t$  is a Parameter

Parametric Form of equation of the line

$$- x = x_0 + at$$

$$- y = y_0 + bt$$

$$- z = z_0 + ct$$

- Symmetric Form of the equation (these equations are the Parametric equation by solving for "t")

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$$

• they are an equal because they have the same "t"

- equation of a Plane

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0 \quad \text{- linear Form}$$

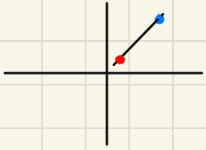
can also be written as

$$ax_0 + by_0 + cz_0 + d = 0 \quad \text{- } d \text{ is a constant}$$

- the distance between the Point  $P_1(x_1, y_1, z_1)$  and the Plane  $ax+by+cz+d=0$  is given by:

$$\frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$$

When you asked to Find the Parametric equation and Symmetric equation and they give you this "the line through the Points  $(-5, 2, 5)$  and  $(1, 6, -2)$ "



You can think of it like something like this  
You take the **Second** Point and subtract it From  
the **First** Point

**Second - First**

$$(1, 6, -2) - (-5, 2, 5)$$

$$\vec{r} \langle 6, 4, -7 \rangle$$

$a, b, c$

• this will  
give you  
your vector

after you found that you can use any point either  $(-5, 2, 5)$  or  $(1, 6, -2)$  as  
 $(x_0, y_0, z_0)$

$$x = x_0 + at = -5 + 6t$$

$$y = y_0 + bt = 2 + 4t$$

$$z = z_0 + ct = 5 - 7t$$

You can solve for  $t$  when you want the symmetric  
equations

Symmetric

$$\frac{x+5}{6} = \frac{y-2}{4} = \frac{z-5}{-7}$$

When a question says "the line through some point  $(2, 1, 0)$  and perpendicular to both  $i+j$  and  $j+k$ " you need to take the cross product of  $\vec{r}_1 \langle i, j, 0 \rangle$  and  $\vec{r}_2 \langle 0, j, k \rangle$  these are 3d so include a zero if it one of them isn't there

$$\vec{r}_1 \times \vec{r}_2 = \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

• the resulting answer will be the actual vector you  
use in the Parametric equations with the given  
point  $P(2, 1, 0)$

If you are given a point as we seen above  $(2, 1, 0)$  but only one  $\vec{r}$  vector and  
the question says Find the line through the point and perpendicular to  $\vec{r}$   
You are already given the  $\vec{r}$  no need to apply cross product you can  
begin to place them into the Parametric equations

if you are given a Problem that Says "Find the Parametric equation and the For the OF inter Section of the Planes  $x+y+z=1$  and  $x-y+z=3$ "

• You can get the  $\vec{r}_1$  and  $\vec{r}_2$  From the equations  $\vec{r}_1 = \langle 1, 1, 1 \rangle$   $\vec{r}_2 = \langle 1, -1, 1 \rangle$  then you apply the Cross Product

$$\vec{r}_1 \times \vec{r}_2 = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$$

$$i \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + k \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$i[(1 \cdot 1) - (1 \cdot 1)] - j[(1 \cdot 1) - (1 \cdot 1)] + k[(1 \cdot -1) - (1 \cdot 1)]$$

$$2i - 0j - 2k = \langle 2, 0, -2 \rangle$$

but now you need to Find the actual Point because it wasn't given you can Pick any number and Set it to either  $x, y, z$

$$\boxed{z=0}$$

$$x+y+(0)=1 \Rightarrow x+y=1 \approx x=1-y$$

$$x-y+(0)=3 \Rightarrow x-y=3$$

$$x-y=3$$

$$(1-y)-y=3$$

$$1-2y=3$$

$$-2y=2$$

$$\boxed{y=-1}$$

$$x=1-y$$

$$x=1-(-1)$$

$$\boxed{x=2}$$

$$P(2, -1, 0)$$

$$\vec{r} = \langle 2, 0, -2 \rangle$$

you can now Plug them into the Parametric equation

When you are given a question that says "Find Parametric equations For the line through  $(2, 4, 6)$  that is Perpendicular to the Plane  $x - y + 3z = 7$ "

You are already given the Point and the  $\vec{r}$

$\vec{r} = \langle 1, -1, 3 \rangle$  • Just Plug them into the Parametric equations

$P = (2, 4, 6)$

$$\begin{array}{lcl} x = x_0 + at & \Rightarrow & 2 + t \\ y = y_0 + bt & \Rightarrow & 4 - t \\ z = z_0 + ct & \Rightarrow & 6 + 3t \end{array}$$

• how to determine if you have Parallel, intersecting or skewed and how to Find the Point of intersection

$$l_1: x = 3 + 2t, y = 4 - t, z = 1 + 3t$$

$$l_2: x = 1 + 4s, y = 3 - 2s, z = 4 + 5s$$

- testing if they are Parallel

if  $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$  are all the same they are Parallel

$$\frac{2}{4} \neq \frac{-1}{-2} \neq \frac{3}{5} \quad \bullet \text{ they are not the same so they aren't Parallel}$$

• testing For intersecting line by Find Point of intersection

$$3 + 2t = 1 + 4s$$

$$2t = -2 + 4s$$

$$t = \frac{-2 + 4s}{2}$$

$$4 - t = 3 - 2s$$

$$4 - \left( \frac{-2 + 4s}{2} \right) = 3 - 2s$$

$$4 - (-1 + 2s) = 3 - 2s$$

$$4 + 1 - 2s = 3 - 2s$$

$$5 - 2s = 3 - 2s$$

We have no solution here which means they do not intersect

So if they aren't Parallel or intersecting they are skewed

$$l_1: x = 5 - 12t, y = 3 + 9t, z = 1 - 3t$$

$$l_2: x = 3 + 8s, y = -6s, z = 4 + 2s$$

testing For Parallel

$$-\frac{12}{8} = \frac{9}{-6} = \frac{-3}{2} \approx -\frac{3}{2} = -\frac{3}{2} = -\frac{3}{2}$$

these are all Similar so they are **Parallel**

★ When you want to Find if there is a Point of intersection you want to solve for  $t$  and  $s$  and plug them back into the equation if they result in  $l_1$  and  $l_2$  to be the same you have a Point of intersection

Finding an equation For the Plane

- equation of a Plane:  $a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$

the Plane through the Point  $(9, -4, 5)$  and Parallel to the Plane  $z = 2x - 3y$

$$2x - 3y - z = 0$$

$$\vec{n} \langle 2, 3, -1 \rangle \quad P(9, -4, 5)$$

Plug these into the equation

$$2(x-9) - 3(y+4) - 1(z+5) = 0$$

$$2x - 18 - 3y - 12 - z - 5 = 0$$

$$\boxed{2x - 3y - z = 35}$$

the Plane through the origin and the Points  $(2, 1, 2)$ ,  $(3, -8, 6)$ , and  $(-2, -3, 1)$

$$P(2, 1, 2) \quad Q(3, -8, 6) \quad R(-2, -3, 1)$$

$$\vec{PQ} = Q - P$$

$$\vec{PR} = R - P$$

• then take the  
Cross Product of  
these two and  
that should give  
you  $\vec{n}$

★ it does not matter what Point you use for  $(x_0, y_0, z_0)$  you can use any of the three provided for you then just plug them into the equation of the Plane

Find Parametric equations for the tangent to the curve with the given Parametric equation

$$x = t^2 + 1, \quad y = 4\sqrt{t}, \quad z = e^{t^2 - t} \quad (2, 4, 1)$$

If you aren't given a  $t$ -value you will be given a Point From that Point you are expected to find the  $t$ -value

$$\begin{aligned} t^2 + 1 &= 2 \\ \sqrt{t^2} &= \sqrt{1} \\ t &= \pm 1 \end{aligned}$$

$$\begin{aligned} 4\sqrt{t} &= 4 \\ (\sqrt{t})^2 &= (1)^2 \\ t &= 1 \end{aligned}$$

$$\begin{aligned} e^{t^2 - t} &= 1 \\ t^2 - t \cdot \ln(e) &= \ln(1) \\ t^2 - t &= 0 \\ t(t-1) &= 0 \\ t=0 \text{ and } t=1 \end{aligned}$$

You should get the same " $t$ " value from all of them but here we do get a number in common which is one you can check if that is the correct answer by plugging it back into the given equation you will know you are correct if you get back the point

$$\begin{aligned} x &= (1)^2 + 1 & y &= 4\sqrt{1} & z &= e^{1^2 - 1} & \cdot \text{we get } (2, 4, 1) \text{ which means we do have} \\ x &= 2 & y &= 4 & z &= 1 & \text{the correct } t\text{-value} \end{aligned}$$

take the derivative of the provided equations then plug in the found  $t$ -value

$$x' = 2t, \quad y' = \frac{2}{\sqrt{t}}, \quad z' = (2t-1) \cdot e^{t^2 - t}$$

$$\vec{r}(2, 2, 1)$$

You can plug this information into the parametric equations

$$\vec{r}(2, 2, 1) \quad P(2, 4, 1)$$

$$\begin{aligned} x &= x_0 + at = 2 + 2t \\ y &= y_0 + bt = 4 + 2t \\ z &= z_0 + ct = 1 + t \end{aligned}$$

the curves  $r_1(t) = \langle t, t^2, t^3 \rangle$  and  $r_2(t) = \langle \sin(t), \sin(2t), t \rangle$  intersect at the origin Find their angle of intersection

You are told that they intersect at the origin which is  $(0,0,0)$  you want to begin finding the  $t$  value you can do this by setting the  $r_1(t)$   $x, y, z$  equal to the  $(0,0,0)$  and solve for  $t$

$$(0,0,0)$$

$$t = 0$$

$$\sqrt{t^2} = \sqrt{0} = 0$$

$$\sqrt[3]{t^3} = \sqrt[3]{0} = 0$$

$$t = 0$$

• You can test if this is correct by plugging the  $t$  you found into  $r_1(t)$  and  $r_2(t)$  and seeing if you get the point

• Now take the derivatives of  $r_1(t)$  and  $r_2(t)$  then plug in the  $t$

$$r_1' \langle 1, 2t, 3t^2 \rangle = r_1' \langle 1, 0, 0 \rangle$$

$$r_2' \langle \cos t, 2\cos(2t), 1 \rangle = r_2' \langle 1, 2, 1 \rangle$$

• You now can use  $\theta = \cos^{-1} \left( \frac{A \cdot B}{|A| \cdot |B|} \right)$

$$\theta = \cos^{-1} \left( \frac{1}{1 \cdot \sqrt{6}} \right) = 66^\circ$$



• Finding the domain of the Vector Function

$$r(t) = \left\langle \frac{1}{t^2 - 4}, \sqrt{t+3}, \ln(t-1) \right\rangle$$

$$t^2 - 4 > 0$$

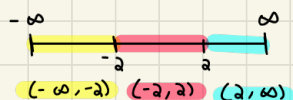
★ this is a non-linear function and needs a bit more steps

Step 1: Solve it like a normal function

$$t^2 - 4 = 0$$

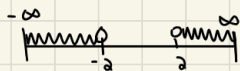
$$\sqrt{t^2} = \sqrt{4} \Rightarrow t = \pm 2$$

Step 2: Plot it on number line and pick points in between



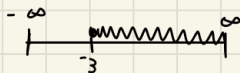
$$\begin{array}{llll} (-\infty, -2) \Rightarrow -3 & (-3)^2 - 4 > 0 & \Rightarrow & 5 > 0 \checkmark \\ (-2, 2) \Rightarrow 0 & (0)^2 - 4 > 0 & \Rightarrow & -4 > 0 \times \\ (2, \infty) \Rightarrow 3 & (3)^2 - 4 > 0 & \Rightarrow & 5 > 0 \checkmark \end{array}$$

• the points that have checkmarks on them are the ones you can use



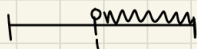
$$t + 3 \geq 0$$

$$t \geq -3$$



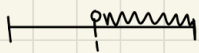
$$t - 1 > 0$$

$$t > 1$$

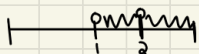


- Now you just combine them all into one (look at the one with the most restrictions)

- For us on this problem the one with the most restrictions is:



From  $t^2 - 4$  there was a "2" that we can add to this



Now we can write our final answer like this:

$$(1, 2) \cup (2, \infty)$$

- Finding the limit of Vector Functions

$$\lim_{t \rightarrow 0} \left( 2te^{-3t} \vec{i} + \frac{t^2}{1 - \cos t} \vec{j} + \sin(2t) \vec{k} \right)$$

$$\lim_{t \rightarrow 0} 2te^{-3t} \vec{i} + \lim_{t \rightarrow 0} \frac{t^2}{1 - \cos t} \vec{j} + \lim_{t \rightarrow 0} \sin(2t) \vec{k}$$

$$\lim_{t \rightarrow 0} 2(0)e^{-3(0)} \vec{i} + \lim_{t \rightarrow 0} \frac{(0)^2}{1 - 1} \vec{j} + \lim_{t \rightarrow 0} \sin(2(0)) \vec{k}$$

$e^{-3(0)} = 1$

- apply L'Hospital's rules

$$0 \vec{i} + \lim_{t \rightarrow 0} \frac{2t}{\sin(t)} \vec{j} + 0 \vec{k}$$

$$0 \vec{i} + \lim_{t \rightarrow 0} \frac{2t}{\sin(t)} \vec{j} + 0 \vec{k}$$

- apply L'Hospital's rule

$$0 \vec{i} + \lim_{t \rightarrow 0} \frac{2}{\cos t} \vec{j} + 0 \vec{k} \Rightarrow 0 \vec{i} + 2 \vec{j} + 0 \vec{k}$$

• When you are told to draw the Projection on a  $xy$ -Plane and given a equation

$\vec{r}(t) = \langle t, t^2, t^4 \rangle$  ignore the  $z$ -value they only want the Projection on the  $xy$ -Plane

| $t$ | $x = t^2$ | $y = t^4$ |
|-----|-----------|-----------|
| -2  | -2        | 4         |
| -1  | -1        | 1         |
| 0   | 0         | 0         |
| 1   | 1         | 1         |
| 2   | 2         | 4         |

• once you got the Points down you just Plot them

When you are given a Problem asking for "the Plane that Passes through the Point  $(3, 5, -1)$  and contains the line  $x = 4 - t, y = 2t, z = -3t$ "

$P_1(3, 5, -1)$  and  $P_2(4, 0, 0)$

From the given Parametric equations

$\langle 1, -5, 1 \rangle$   $\langle -1, 2, -3 \rangle$

this was gotten by subtracting

$P_2 - P_1$

take the Cross product of the two Vectors

$$\begin{vmatrix} i & j & k \\ 1 & -5 & 1 \\ -1 & 2 & -3 \end{vmatrix}$$

$$i \begin{vmatrix} -5 & 1 \\ 2 & -3 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ -1 & -3 \end{vmatrix} + k \begin{vmatrix} 1 & -5 \\ -1 & 2 \end{vmatrix}$$

$13i + 2j - 3k$

$\vec{n} = \langle 13, 2, -3 \rangle$

$P(3, 5, -1)$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$13(x - 3) + 2(y - 5) - 3(z + 1) = 0$$

Find the equation of the Plane that contains the curve with the given vector equation

$$\langle t, t^2, 2t \rangle$$

★ remember the equation of the line is linear

$$ax + by + cz = d$$

$$\begin{aligned}x &= t \\y &= t^2 \\z &= 2t\end{aligned}$$

$$\frac{xy}{y=x^2}$$

$$\frac{yz}{\boxed{z=2x}} \quad \frac{xz}{}$$

★ the whole point of this is to substitute and get a linear equation

### • Derivatives and integrals of Vector Functions

Sketch the Plane curve with the given vector equation Find  $\vec{r}(t)$  and sketch the Position Vector  $\vec{r}(t)$  and  $\vec{r}'(t)$  at the given  $t$ -Value

a)  $\vec{r}(t) = \langle t-1, t^2+1 \rangle$  at  $t=2$

| t  | t-1 | t <sup>2</sup> +1 |
|----|-----|-------------------|
| -2 | -3  | 5                 |
| -1 | -2  | 2                 |
| 0  | -1  | 1                 |
| 1  | 0   | 2                 |
| 2  | 1   | 10                |



• do not place arrows at the end the arrows here are representing movement from the left to the right

• Find the derivative

$$\vec{r}(t) = \langle t-1, t^2+1 \rangle \text{ at } t=2$$

$$\vec{r}'(t) = \langle 1, 2t \rangle$$

Find the position from  $\vec{r}(t)$  and  $\vec{r}'(t)$  by plugging in  $t=2$  and graph it

$$\vec{r} \langle 1, 5 \rangle$$

$$\vec{r}' \langle 1, 4 \rangle$$



Find the derivative of the following vector valued function (derivative rule still apply, noting changes you will apply all derivative rules if they are needed)

$$\vec{r}(t) = \langle e^{2t}, t^2 \ln t, \frac{t^2}{2t+1} \rangle$$

•  $e^{2t}$ :

apply chain rule

$$= 2e^{2t}$$

•  $t^2 \ln t$ :

apply product rule

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = t^2$$

$$g = \ln t$$

$$\frac{df}{dx} = 2t$$

$$\frac{dg}{dx} = \frac{1}{t}$$

$$(2t)(\ln t) + (t^2)\left(\frac{1}{t}\right)$$

•  $\frac{t^2}{2t+1}$ : apply product rule

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{F^2}$$

$$f = t^2$$

$$g = 2t+1$$

$$\frac{df}{dx} = 2t$$

$$\frac{dg}{dx} = 2$$

$$= \frac{2t^2 + 2t}{(2t+1)^2}$$

$$\vec{r}(t) = \left\langle 2e^{2t}, 2t \ln t + t, \frac{2t^2 + 2t}{(2t+1)^2} \right\rangle$$

• Find the unit tangent vector  $\vec{T}(t)$  at the given point

equation for tangent vector  $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$

a)  $\vec{r}(t) = \langle \tan^{-1}(t), \sin(\pi t), 4t^3 - 3t^2 \rangle$  at  $t=1$

$$\vec{r}'(t) = \left\langle \frac{1}{1+t^2}, \pi \cos(\pi t), 12t^2 - 6t \right\rangle$$

• began by taking the derivative first then plug in the given "t" value

$$\vec{r}'(1) = \left\langle \frac{1}{2}, -\pi, 6 \right\rangle$$

$$\sqrt{\left(\frac{1}{2}\right)^2 + (-\pi)^2 + (6)^2} = \sqrt{\pi^2 + \frac{145}{4}} = \frac{\sqrt{4\pi^2 + 145}}{2}$$

$$\vec{T}(t) = \left\langle \frac{\frac{1}{2}}{\frac{\sqrt{4\pi^2 + 145}}{2}}, \frac{-\pi}{\frac{\sqrt{4\pi^2 + 145}}{2}}, \frac{6}{\frac{\sqrt{4\pi^2 + 145}}{2}} \right\rangle$$

• apply keep change flip

$$= \left\langle \frac{1}{\sqrt{4\pi^2 + 145}}, \frac{-2\pi}{\sqrt{4\pi^2 + 145}}, \frac{12}{\sqrt{4\pi^2 + 145}} \right\rangle$$

$$\vec{r}(t) = \langle 5 \sin t, t, 5 \cos t \rangle$$

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

$$\vec{r}'(t) = \langle 5 \cos t, 1, -5 \sin t \rangle$$

$$= \frac{1}{\sqrt{26}} \langle 5 \cos t, 1, -5 \sin t \rangle$$

$$\begin{aligned} |\vec{r}'(t)| &= \sqrt{(5 \cos t)^2 + (1)^2 + (-5 \sin t)^2} \\ &= \sqrt{25 \cos^2 t + 25 \sin^2 t + 1} \\ &= \sqrt{25 (\cos^2 t + \sin^2 t) + 1} \\ &= \sqrt{25(1) + 1} = \sqrt{26} \end{aligned}$$

unit normal vector

$$\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}$$

$$\vec{T}'(t) = \left\langle \frac{\frac{5}{\sqrt{26}} (-\sin t)}{\frac{5}{\sqrt{26}}}, 0, \frac{\frac{-5}{\sqrt{26}} (\cos t)}{\frac{5}{\sqrt{26}}} \right\rangle$$

$$\vec{T}'(t) = \left\langle \frac{5}{\sqrt{26}} (-\sin t), 0, \frac{-5}{\sqrt{26}} (\cos t) \right\rangle$$

$$\begin{aligned} |\vec{T}'(t)| &= \sqrt{\left(\frac{5}{\sqrt{26}} \sin t\right)^2 + (0)^2 + \left(\frac{5}{\sqrt{26}} \cos t\right)^2} \\ &= \sqrt{\frac{25}{26} (\sin^2 t + \cos^2 t)} \\ &= \sqrt{\frac{25}{26} (1)} = \frac{5}{\sqrt{26}} \end{aligned}$$

$$\langle -\sin t, 0, \cos t \rangle$$

- binormal Vector

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$$

$$\vec{T}(t) = \frac{1}{\sqrt{26}} \langle 5 \cos(t), 1, -5 \sin(t) \rangle$$

$$\vec{N}(t) = \langle -\sin(t), 0, \cos(t) \rangle$$

- $\vec{B}(t)$  is Perpendicular to  $\vec{T}(t)$  and  $\vec{N}(t)$
- You need to Find the cross Product and you will be dealing with variables but you aren't doing anything special as you are doing is cross Product

- 
- Curvature of a curve can be found by applying any of these equation

$$K = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} \quad , \quad K = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}$$

- You can apply any one

- ↗
- We will apply this because we already have the components and all we have to do is plug in

$$|\vec{T}'(t)| = \frac{5}{\sqrt{26}} \quad |\vec{r}'(t)| = \sqrt{26}$$

$$K = \frac{\frac{5}{\sqrt{26}}}{\sqrt{26}} = \frac{5}{\sqrt{26}} \cdot \frac{1}{\sqrt{26}} = \frac{5}{26}$$

- You can use the other equation but you will do cross Product with variables

### • Special Case

- When you are given a two dimensional problem and told to find the curvature you need to apply this equation:

$$K = \frac{|f''(x)|}{[1 + (f'(x))^2]^{\frac{3}{2}}}$$

- example of two dimensional equation

$$y = xe^x$$

- arch length

- arch length of a space curve  $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$  for  $a \leq t \leq b$  is given by:

$$\int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}$$

You began by taking the derivative once you did that you square it  
Place it in a square root then take the integral with the given points